

L1.8 Linearization & Taylor

Recall: L1.7b Related Rates

Name _____

Solo – 9 minutes

1. Differentiate each with respect to t .

(a) $A = \pi r^2$

(b) $V = \frac{1}{3}\pi r^2 h$

(c) Suppose $\dot{V}$ is known and you want $\dot{h}$. Why can't you solve 1(b) for it? What would you need before differentiating?

2. Oil spreads in a circle on still water. The radius grows at 0.5 m/min. How fast is the area growing when the radius is 20 m?

Use the template.

3. Now the slick has stopped spreading. Its radius is not changing at all — it is simply **measured**, as 20 ± 0.1 m.

What can you say about its area? Write down what you think the calculation should be.